

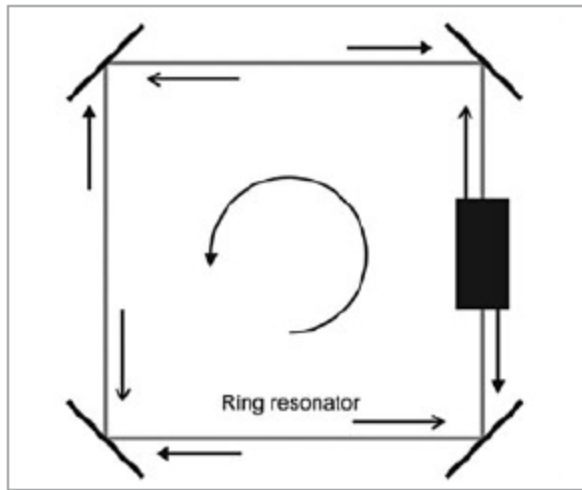


Fig. 1: Square-type ring laser geometry



Fig. 5: Lasing triangular ring laser cavity

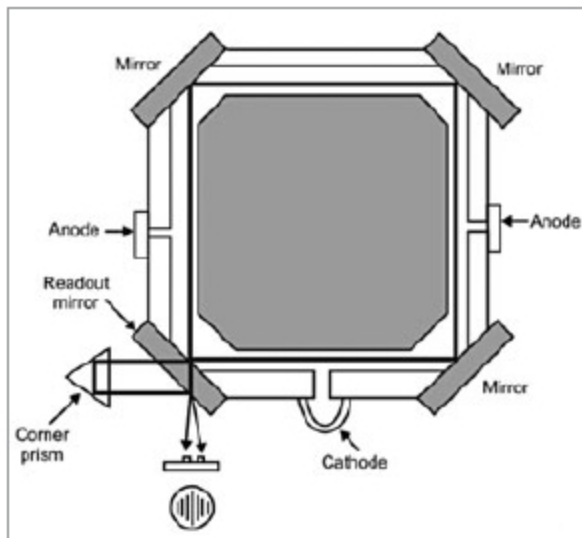


Fig. 2: Constructional features of square geometry RLG

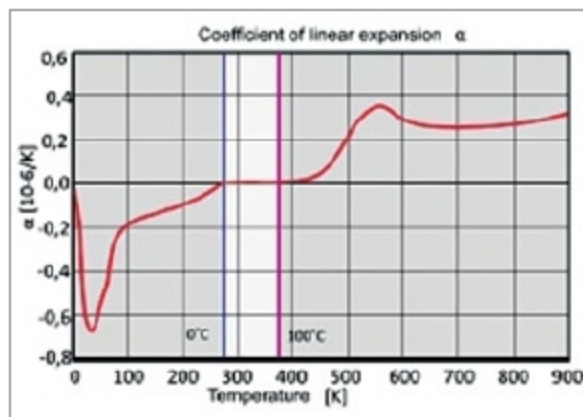


Fig. 6: Temperature coefficient of linear expansion of ZERODUR

Frequency difference produced as a result of rotation is obtained by the interference of the two beams. The two beams are usually combined with the help of a corner prism, as shown in Fig. 1.

Figs 2 and 3 bring out the constructional features of the square-type ring laser geometry.

Another commonly used ring laser cavity structure is triangular geometry, as shown in Fig. 4. This type of geometry is popular with RLGs from Honeywell. Fig. 5 shows a lasing triangular ring laser cavity.

Going back to Sagnac equation, constancy of A and P is important to achieve the kind of angular resolution present-day RLGs are capable of. It is due to this that the gyroscope cavity material should have near-zero thermal expansion coefficient.

ZERODUR is one such material. It is a registered trade name of a near-zero thermal expansion transparent glass ceramic made by Schott, Germany. The material has a near-zero thermal expansion coefficient over a temperature range of 0°C to 100°C (Fig. 6) and transmission of 90 per cent in the spectral range that extends from 400nm to greater than 2000nm (Fig. 7).

RLGs suffer from lock-in effect at very



Fig. 3: Square geometry RLG

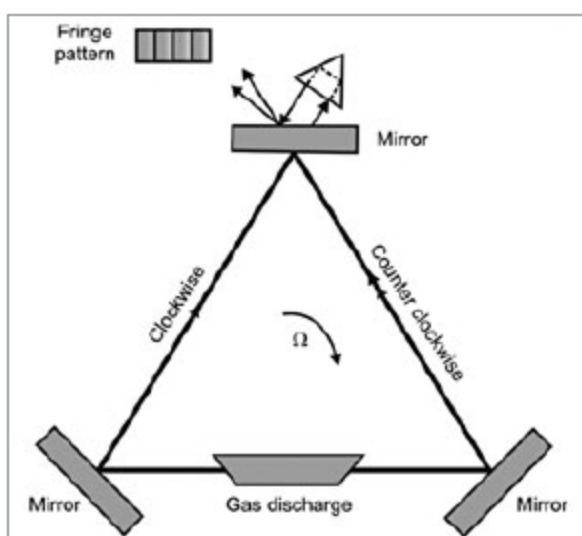


Fig. 4: Triangular geometry RLG

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Plot No. EL-60, MIDC Electronic Zone,
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Navi Mumbai - 400710 (INDIA)
Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
Fax : 0091-22-27673310, 27673330
Email : sales@mecoinst.com
Web : www.mecoinst.com